

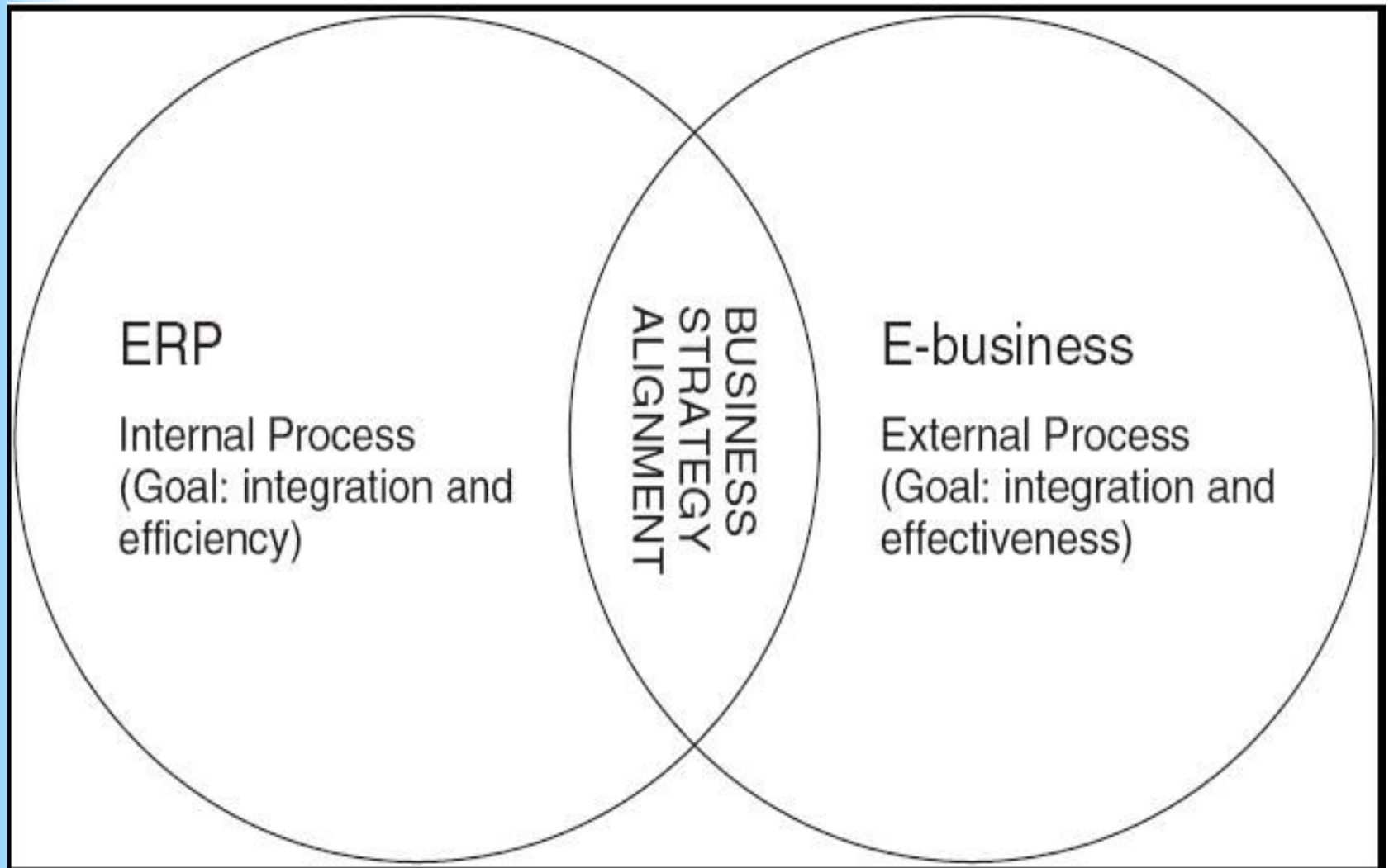
Week 2 - Continuation

Introduction to Enterprise Systems for Management

E-Business and ERP

E-Business	ERP
Focuses on linking a business with its external partners and stakeholders	Focuses on integrating the internal functional silos of the organization into an enterprise application
Disruptive technology—Totally transformed the way a business operates in terms of buying and selling, customer service, and relationships with suppliers	Adaptive technology—Merged the early data processing and integration efforts within an organization
Early focus of e-Business was on communication (e-mail), collaboration (calendar, scheduling, group support), marketing and promotion (Web sites), and E-commerce (<i>Front office functions</i>)	Focus of ERP systems was mainly on data sharing, systems integration, business process change, and improving decision making through the access of data from a single source (<i>Back office functions</i>)

Figure 1-8 e-Business and ERP



System Benefits of an ERP System

- Integration of data and applications across functional areas (i.e., data can be entered once and used by all applications; thus improving accuracy and quality of the data).
- Improvements in maintenance and support as IT staff is centralized.
- Consistency of the user interface across various applications means less employee training, better productivity, and cross-functional job movements.
- Security of data and applications is enhanced due to better controls and centralization of hardware.

System Limitations of an ERP System

- Complexity of installing, configuring, and maintaining the system increases, thus requiring specialized IT staff, hardware, and network facilities.
- Consolidation of IT hardware, software, and people resources can be cumbersome and difficult to attain.
- Data conversion and transformation from an old system to a new one can be a tedious and complex process.
- Retraining of IT staff and end users of the new system can produce resistance and reduce productivity.

Business Benefits of an ERP System

- Increasing agility of the organization in terms of responding to changes in environment for growth and maintaining market share.
- Information sharing helps collaboration between units.
- Linking and exchanging information in real-time with supply-chain partners improves efficiency.
- Better customer service due to quicker information flow across departments.
- Efficiency of business processes are enhanced due to the re-engineering of business processes.

Business Limitations of an ERP System

- Retraining of all employees with the new system can be costly and time consuming.
- Change of business roles and department boundaries can create upheaval and resistance to the new system.

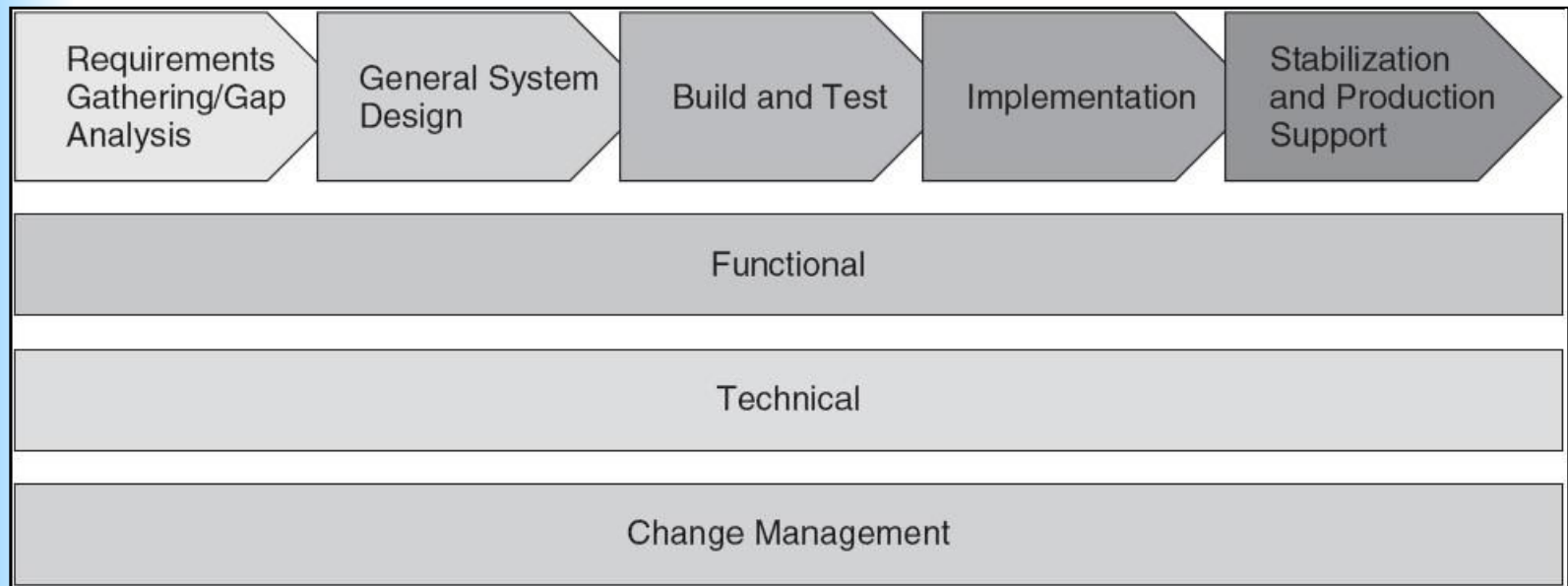
ERP Implementation (Business Process Management)

- Business process management is the understanding, visibility, and control of business processes.
- BPM has a prescribed methodology that should be followed to help document business processes and understand their use throughout the business.
- Improved business processes may result in:
 - Improved customer satisfaction.
 - Reductions in cost.
 - Increased productivity by allocating resources to more value-added activities.

ERP Implementation (ERP Life Cycle)

- The key to a successful implementation is to use a proven methodology, take it one step at a time, and begin with the planning and understanding the ERP life cycle.
- ERP system implementations are very risky, and using a well-defined project plan with a proven methodology will assist in managing those risks.
- There must be a strong well-communicated need to make the change from the existing information systems/applications to an ERP system.

Figure 1-10 ERP Implementation Methodology



Software and Vendor Selection

- It is best for an organization that does not have the experience in developing ERP systems to purchase one on the market.
- Before selecting a vendor, the organization must carefully evaluate its current and future needs in enterprise management systems.
- Review the organization's existing hardware, network, and software infrastructure, and the resources available for the implementation.

Vendor Evaluation

- Business functions or modules supported by their software.
- Features and integration capabilities of the software.
- Financial viability of the vendor as well as length of time they have been in business.
- Licensing and upgrade policies.
- Customer service and help desk support.

Vendor Evaluation (Cont'd)

- Total cost of ownership.
- IT infrastructure requirements.
- Third-party software integration.
- Legacy systems support and integration.
- Consulting and training services.
- Future goals and plans for the short and long term.

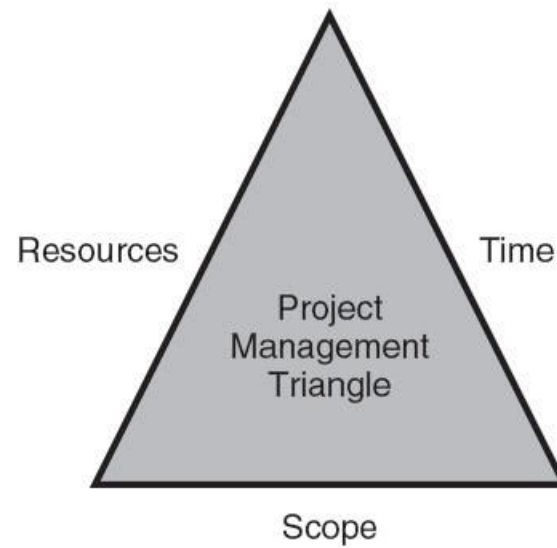
Operations and Post-Implementation

- Going live (“Go-live”) is one of the most critical points in a project’s success.
- It is vital to focus the efforts of all project teams to ensure that task and activities are completed before going live.
 - This allows project management to address any outstanding issues that may jeopardize the Go-live date.
 - This involves a readiness process that needs to include as many team members and appropriate users and managers as possible.

Five Areas of Stabilization are Important:

1. Training for end-users.
2. Reactive support (i.e., help desk for troubleshooting).
3. Auditing support to make sure data quality is not compromised by new system.
4. Data fix to resolve data migration and errors revealed by audits.
5. New features and functionalities to support the evolving needs of the organization.

Figure 1-12 Project Management



People and Organization

Project Management

For an ERP system to be implemented successfully, project management must provide strong leadership, a clear and understood implementation plan, and close monitoring of the budget.

Consultants

It is often the case for organizations without much ERP implementation experience to use implementation partners such as consultants.

People and Organization (Cont'd)

Change Management

This helps prepare for changes to how business is done. In implementing new systems, communicating, preparing, and setting expectations is as important as providing training and support.

Business Process Re-engineering

Business processes will need to be changed, adjusted, or adapted to the new system to use the functionality of an ERP system fully.

Global, Ethical and Security Management

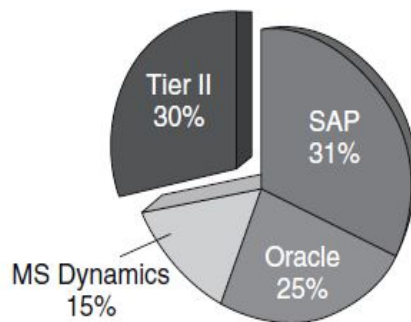
Outsourcing overseas, ethical issues, and problems with system security have also attracted a lot of attention in ERP implementation.

ERP Market Tiers

Tier I, Tier II, and Tier III ERP Software Vendors

Sample Vendors

Tier I	Tier II	Tier III
SAP	Epicor	ABAS
Oracle	Sage	Activant Solutions Inc.
Oracle—e-Business Suite	Infor	Bowen and Groves
Oracle—JD Edwards	IFS	Compiere
Oracle—Peoplesoft	QAD	Exact
Microsoft Dynamics	Lawson	NetSuite
	CDC Software	Visibility
		CGS
		Hansa World
		Consona
		Syspro



ERP Vendors

SAP

SAP is the recognized global leader among ERP vendors with over 12 million users. Its solutions are for all types of industries and for every major market. www.sap.com

Oracle/Peoplesoft

As the second largest ERP vendor, Oracle provides solutions divided by industry category and promises long-term support for customers of PeopleSoft- (acquired in 2004). www.oracle.com

Infor

The world's third largest provider of enterprise software. It delivers integrated enterprise solutions in supply chain, customer relationship and suppliers management.

ERP Vendors (Cont'd)

Microsoft Dynamics

Formerly Microsoft Business Solutions or Great Plains, Microsoft Dynamics is a comprehensive business-management solution built on the Microsoft platform.

Microsoft Dynamics integrates finances, e-commerce, supply chain, manufacturing, project accounting, field service, customer relationships, and human resources.

Lawson

Industry-tailored software solutions that include enterprise performance management, distribution, financials, human resources, procurement, and retail operations. www.Lawson.com

ERP Vendors (Cont'd)

SSA Global

Acquired Baan in 2004. They claim to offer solutions that accomplish specific goals in shorter time frames and are more efficient with time.

Epicor

This company provides enterprise software solutions for midmarket companies around the world. Claims to have solutions to a variety of needs, whether a customer is looking for a complete end-to-end enterprise software solution or a specific application.

Software Extensions and Trends

- As e-Business firms started growing bigger with advanced needs in HR, accounting, and warehousing, non-ERP vendors were unable to support their requirements.
- ERP vendors were starting to expand their functionality to the Internet and *e-Business*.
- Intense competition and fluctuating sales have forced the ERP vendors to expand functionality to add value.
- The saturation of the demand in big business and the lucrative nature of the small and mid-sized business markets have led vendors like SAP and Oracle to enter the small business market.

Software Extensions and Trends

- SOA implementation will continue to grow as a factor in ERP purchase decisions because vendors are using creative marketing around product strategies versus buying what is currently available.
- Another shift is toward recurring and variable revenue models - Maintenance charges driving industry growth.
- The other major revenue shift is toward software as a service or hosted subscription-based applications.
- Social networking and open-source software solutions are also poised for significant growth.

Implications for Management

- ERP systems implementation is a complex organizational activity.
 - Important to evaluate and learn from the successes and failures.
 - Managing risk is all about keeping project focus and clear communications throughout the organization.
- ERP systems implementation requires strong project management oversight.
- ERP systems provide improved and added functionality for an organization.
- ERP systems are set to proliferate globally.

Summary

- Whereas the risks for implementing an ERP are greater, the payoff is very high for organizations.
- The integration of data helps an organization to better meet the demands of a fast and dynamic business world.
- The use of ERP systems provides for integrated data and business processes, thereby creating opportunities for organizations to expand and change as their business changes.

Summary (Cont'd)

- ERP components consist of hardware, software, information, process, and people to perform the fundamental phases of an information system: input, process, and output.
- ERP system architecture is a blueprint of the actual ERP system. There are two types of architecture: physical and logical.
- The selection of a system must be based on the needs of the organization and how well a vendor meets those needs now or in the future.

Summary (Cont'd)

- To be successful in implementing an ERP system, an organization and its management must clearly understand the implementation process.
 - The key to this is the application of an ERP life cycle and methodology throughout an implementation.
- People and organizations are an important part of the implementation process. Without in-house experts, either the software vendor or a third party should be hired and used to assist or lead the project.
- Whereas ERP implementations are costly in time and resources, the greater costs are in process change, system maintenance, and remaining current.